COURSE 4 · CHEMISTRY

Chemistry

Printable flashcards

153 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 4.1 · Matter and atomic structure

EQUATION Write the equation: Mass number	$A = Z + N$
EQUATION Write the equation: Relative atomic mass	$A_r = \sum(\text{isotope mass} \times \text{fractional abundance})$
EQUATION Write the equation: Sodium's configuration	$\text{Na}: \ 1s^2\,2s^2\,2p^6\,3s^1 \quad (2, 8, 1)$
KEY IDEA Complete the key idea: Z = protons (defines …	Z = protons (defines the element); A = protons + neutrons
KEY IDEA Complete the key idea: Isotopes differ …	Isotopes differ in neutron number only
KEY IDEA Complete the key idea: Valence electrons …	Valence electrons decide chemistry
KEY IDEA Complete the key idea: Shells fill 2, 8, …	Shells fill 2, 8, 8 for the first 20 elements
QUICK CHECK The nucleus contains	protons and neutrons
QUICK CHECK An atom with 8 protons and 10 electrons is	a 2− ion
QUICK CHECK Mass number is	protons + neutrons
QUICK CHECK Maximum electrons in the first shell:	2

PROBLEM How many neutrons are in an atom of iron-56 (Z = 26)?	30. N = A − Z = 56 − 26.
PROBLEM What defines which element an atom is?	number of protons. The atomic number, Z, the proton count.
PROBLEM How many electrons does a Mg^{2+} ion have? (Mg: Z = 12)	10. 2+ means two lost: 10.
PROBLEM Give the shell arrangement of chlorine (Z = 17).	2, 8, 7. Fill 2, then 8, then the remaining 7.
PROBLEM Boron is 20% boron-10 and 80% boron-11. What is its relative atomic mass?	10.8. = 10.8.
PROBLEM Carbon-12 and carbon-14 differ in	neutrons. Isotopes differ only in neutron number.
PROBLEM How many valence electrons does sulfur (Z = 16) have?	6. Outer shell: 6.
PROBLEM Explain why isotopes of the same element have almost identical chemistry.	Chemistry is decided by electrons, especially the valence electrons. Isotopes have the same number of protons and therefore the same number and arrangement of electrons. Only the neutron count differs, which changes the mass slightly but not how the atom bonds.

Module 4.2 · The periodic table

EQUATION Write the equation: Ionisation (first)	$\mathrm{X(g) \rightarrow X^{+}(g) + e^{-}}$
EQUATION Write the equation: Alkali metal with water	$\mathrm{2Na + 2H_2O \rightarrow 2NaOH + H_2}$
KEY IDEA Complete the key idea: Group = valence …	Group = valence electrons = similar chemistry

KEY IDEA Complete the key idea: Period = …	Period = number of shells
KEY IDEA Complete the key idea: Across a period: smaller …	Across a period: smaller atoms, higher ionisation energy and electronegativity
KEY IDEA Complete the key idea: Down a group: larger …	Down a group: larger atoms, outer electrons easier to remove
QUICK CHECK Periods are	rows
QUICK CHECK Electronegativity increases	across a period
QUICK CHECK Mendeleev's table was confirmed when	predicted elements were found
QUICK CHECK Transition metals often	form coloured compounds
PROBLEM Elements in the same group have the same number of	valence electrons. That is why they react similarly.
PROBLEM Which group contains the noble gases?	18. Group 18, full outer shells.
PROBLEM Which atom is larger?	sodium. Same period; Cl has more protons pulling the same shell in.
PROBLEM Which is the most reactive alkali metal of these?	potassium. Reactivity increases down group 1.
PROBLEM Which is the most reactive halogen of these?	fluorine. Reactivity decreases down group 17.
PROBLEM Calcium is in group 2. What charge does its ion usually carry? (enter as a number, e.g. 2 for 2+)	2. Two valence electrons, lost to give Ca^{2+}.

PROBLEM Why does ionisation energy rise across period 3?	more protons pulling the same shell. Nuclear charge increases while shielding stays similar.
PROBLEM Using electron arrangements, explain why sodium and chlorine react so readily with each other.	Sodium (2,8,1) has one valence electron that is easy to remove, leaving a full shell. Chlorine (2,8,7) needs one electron to fill its outer shell. Transferring an electron gives Na^+ and Cl^-, both with full shells, and the oppositely charged ions attract strongly, releasing energy.

Module 4.3 · Bonding and molecular shape

EQUATION Write the equation: Ionic formation	$\mathrm{Na} + \mathrm{Cl} \rightarrow \mathrm{Na}^+ + \mathrm{Cl}^-$
EQUATION Write the equation: Covalent sharing in water	$\mathrm{H{-}O{-}H}$
KEY IDEA Complete the key idea: Ionic: transfer, lattices. …	Ionic: transfer, lattices. Covalent: sharing. Metallic: electron sea
KEY IDEA Complete the key idea: VSEPR: electron pairs spread …	VSEPR: electron pairs spread out; lone pairs squeeze angles
KEY IDEA Complete the key idea: Shape decides …	Shape decides whether polar bonds cancel
KEY IDEA Complete the key idea: Hydrogen bonds …	Hydrogen bonds explain water's unusual properties
QUICK CHECK A covalent bond is	a shared pair of electrons
QUICK CHECK Water's shape is	bent
QUICK CHECK The strongest intermolecular force listed is	hydrogen bonding
QUICK CHECK Lone pairs make bond angles	smaller

PROBLEM What type of bonding is in magnesium oxide, MgO?	ionic. Metal plus non-metal: electron transfer.
PROBLEM Why do metals conduct electricity?	they have a sea of free electrons. Delocalised electrons can move.
PROBLEM What is the shape of methane, CH_4?	tetrahedral. Four bonding pairs, no lone pairs.
PROBLEM CO_2 has polar bonds. Why is the molecule non-polar?	it is linear, so the dipoles cancel. Equal and opposite bond dipoles cancel.
PROBLEM How many electrons are shared in a double bond?	4. Each bond is a pair; two bonds = 4 electrons.
PROBLEM Which molecule can form hydrogen bonds with itself?	NH_3. H attached to N, with a lone pair on N.
PROBLEM Sodium chloride conducts electricity	when molten or dissolved. Ions must be free to move.
PROBLEM Explain why ice floats on water.	In ice, hydrogen bonds lock water molecules into an open hexagonal lattice with lots of space. In liquid water the hydrogen bonds keep breaking and reforming, so molecules pack closer together. Ice is therefore less dense than liquid water and floats.

Module 4.4 · The mole and stoichiometry

EQUATION Write the equation: Moles from mass	$n = \frac{m}{M}$
EQUATION Write the equation: Number of particles	$N = nN_A, \quad N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$
EQUATION Write the equation: Combustion of methane	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

EQUATION Write the equation: Percentage yield	$\%\ \text{yield} = \frac{\text{actual}}{\text{theoretical}} \times 100$
EQUATION Write the equation: Concentration and dilution	$c = \frac{n}{V}, \qquad c_1 V_1 = c_2 V_2$
KEY IDEA Complete the key idea: n = m/M; …	n = m/M; particles = n × N_a
KEY IDEA Complete the key idea: Balance with …	Balance with coefficients, never subscripts
KEY IDEA Complete the key idea: Coefficients are …	Coefficients are mole ratios
KEY IDEA Complete the key idea: c = n/V; …	c = n/V; dilution keeps $c_1V_1 = c_2V_2$
QUICK CHECK One mole contains	6.02×10^{23} particles
QUICK CHECK To balance an equation you change	coefficients
QUICK CHECK The limiting reagent	runs out first
QUICK CHECK Molarity is	moles per litre
PROBLEM How many moles are in 44 g of carbon dioxide (M = 44 g/mol)?	1 mol. n = m/M = 44/44.
PROBLEM What is the molar mass of sulfuric acid, H_2SO_4? (H = 1, S = 32, O = 16)	98 g/mol. 2(1) + 32 + 4(16) = 98.
PROBLEM How many molecules are in 9.0 g of water (M = 18 g/mol)?	3.010e+23. N = 0.5 × 6.02 × 10^{23} = 3.0 × 10^{23}.

PROBLEM Balance: _ H_2 + O_2 → _ H_2O. What coefficient goes in front of H_2?	2. $2H_2 + O_2 \rightarrow 2H_2O$ balances 4 H and 2 O.
PROBLEM How many grams of oxygen are needed to burn 8.0 g of methane completely? (CH_4 = 16, O_2 = 32)	32 g. Needs 2 × 0.5 = 1.0 mol O_2 = 32 g.
PROBLEM What is the concentration of 0.25 mol of NaCl dissolved to make 500 mL of solution?	0.5 mol/L. c = n/V = 0.25/0.500.
PROBLEM You need 100 mL of 0.20 M solution from a 1.0 M stock. What volume of stock do you use?	20 mL. $c_1V_1 = c_2V_2$: 1.0 × V_1 = 0.20 × 100.

Module 4.5 · States of matter and gas laws

EQUATION Write the equation: Boyle's law	$P_1V_1 = P_2V_2$
EQUATION Write the equation: Charles' law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
EQUATION Write the equation: Ideal gas law	$PV = nRT, \quad R = 8.314\ \mathrm{J\,mol^{-1}K^{-1}}$
EQUATION Write the equation: Mean kinetic energy	$\overline{E_k} = \frac{3}{2}k_BT$
KEY IDEA Complete the key idea: Particle spacing and …	Particle spacing and motion distinguish solid, liquid, gas
KEY IDEA Complete the key idea: Boyle: PV constant. …	Boyle: PV constant. Charles: V/T constant (kelvin!)
KEY IDEA Complete the key idea: Ideal gas …	Ideal gas law: PV = nRT
KEY IDEA Complete the key idea: Temperature measures …	Temperature measures average particle kinetic energy

QUICK CHECK At constant temperature, halving the volume of a gas	doubles pressure
QUICK CHECK Gas laws require temperature in	K
QUICK CHECK In PV = nRT, n is	moles
QUICK CHECK Particles in a liquid	slide past each other
PROBLEM A gas at 100 kPa occupies 6.0 L. It is compressed to 2.0 L at constant temperature. What is the new pressure?	300 kPa. $100 \times 6 = P_2 \times 2$.
PROBLEM A balloon of 2.0 L at 300 K is heated to 450 K at constant pressure. What is its new volume?	3 L. $V_2 = V_1\,T_2/T_1 = 2 \times 450/300$.
PROBLEM What volume does 2.0 mol of gas occupy at 300 K and 100 000 Pa? Give your answer in litres.	49.88 L. = 0.0499 m^3 ≈ 49.9 L.
PROBLEM How many moles are in a 10 L cylinder at 500 kPa and 290 K?	2.074 mol. ≈ 2.07 mol.
PROBLEM Why must temperatures be in kelvin for gas laws?	the laws are proportional relationships measured from absolute zero. Doubling kelvin doubles volume; doubling °C does not.
PROBLEM According to kinetic theory, heating a gas in a rigid container raises its pressure because	particles move faster and hit the walls harder and more often. Faster particles, more forceful and frequent collisions.
PROBLEM A tyre at 15 °C has a pressure of 240 kPa. After driving it is at 45 °C. Assuming fixed volume, what is the pressure?	265 kPa. 240 × 318/288 ≈ 265 kPa.

PROBLEM Why do real gases deviate from the ideal gas law at high pressure and low temperature?	The ideal law assumes particles have no volume and no attractions. At high pressure, particles are packed close enough that their own volume is a significant fraction of the space. At low temperature, they move slowly enough that attractions between them matter. Both effects make real gases depart from PV = nRT, and eventually they condense.

Module 4.6 · Energy, rates and equilibrium

EQUATION Write the equation: Enthalpy change	$\Delta H = \sum E_{\text{bonds broken}} - \sum E_{\text{bonds made}}$
EQUATION Write the equation: Equilibrium constant for aA + bB ⇌ cC + dD	$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$
EQUATION Write the equation: Haber process	$\mathrm{N_2} + 3\mathrm{H_2} \rightleftharpoons 2\mathrm{NH_3}, \quad \Delta H < 0$
KEY IDEA Complete the key idea: Exothermic: ΔH negative, …	Exothermic: ΔH negative, heat released. Endothermic: ΔH positive
KEY IDEA Complete the key idea: Rate rises with concentration, …	Rate rises with concentration, temperature, surface area and catalysts
KEY IDEA Complete the key idea: Equilibrium is dynamic: …	Equilibrium is dynamic: equal forward and reverse rates
KEY IDEA Complete the key idea: Le Chatelier: the system …	Le Chatelier: the system shifts to oppose a change
QUICK CHECK At equilibrium, forward and reverse rates are	equal
QUICK CHECK Raising temperature generally makes reactions	faster
QUICK CHECK A large K means	mostly products

QUICK CHECK Endothermic reactions have ΔH	positive
PROBLEM An exothermic reaction	releases heat. Exo = out.
PROBLEM A catalyst works by	lowering the activation energy. It provides an easier pathway.
PROBLEM Breaking bonds in a reaction needs 1370 kJ; making the new bonds releases 1652 kJ. What is ΔH?	-282 kJ. ΔH = broken − made = 1370 − 1652.
PROBLEM Powdered marble reacts faster with acid than marble chips because	it has a larger surface area. More exposed particles, more collisions.
PROBLEM For $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic), raising the temperature shifts the equilibrium	left, less ammonia. Shifts in the endothermic direction, which is the reverse.
PROBLEM For the same reaction, increasing pressure shifts it	right. 4 gas molecules on the left, 2 on the right.
PROBLEM For $A \rightleftharpoons B$, at equilibrium [A] = 0.20 M and [B] = 0.80 M. What is K?	4. K = [B]/[A] = 0.80/0.20.
PROBLEM Explain what 'dynamic' means in dynamic equilibrium.	At equilibrium the forward and reverse reactions are both still happening, at equal rates. Molecules are constantly converting in both directions, so on a molecular level the system is busy, but the overall concentrations stay constant.

Module 4.7 · Acids, bases and redox

EQUATION Write the equation: pH	$\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$
EQUATION Write the equation: Neutralisation	$\mathrm{HCl} + \mathrm{NaOH} \rightarrow \mathrm{NaCl} + \mathrm{H_2O}$
EQUATION Write the equation: A redox pair	$\mathrm{Zn} + \mathrm{Cu}^{2+} \rightarrow \mathrm{Zn}^{2+} + \mathrm{Cu}$
EQUATION Write the equation: Water's ion product	$[\mathrm{H}^+][\mathrm{OH}^-] = 10^{-14}$ at $25\ ^\circ\mathrm{C}$
KEY IDEA Complete the key idea: Acids donate …	Acids donate protons; bases accept them
KEY IDEA Complete the key idea: pH = −log[H⁺]; …	pH = −log[H⁺]; each unit is ×10
KEY IDEA Complete the key idea: Acid + base …	Acid + base → salt + water
KEY IDEA Complete the key idea: OIL RIG: oxidation is …	OIL RIG: oxidation is loss, reduction is gain of electrons
QUICK CHECK An acid is a proton	donor
QUICK CHECK pH 2 compared with pH 4 is	100 times as acidic
QUICK CHECK Reduction is	gain of electrons
QUICK CHECK Neutral pH at 25 °C is	7
PROBLEM What is the pH of a solution with [H⁺] = 10^{-4} M?	4. pH = −log(10^{-4}) = 4.

PROBLEM What is $[H^+]$ in mol/L for a solution of pH 3? Give a decimal.	0.001 mol/L. $[H^+] = 10^{-3}$.
PROBLEM In Zn + $Cu^{2+} \rightarrow Zn^{2+}$ + Cu, which species is oxidised?	Zn. Zn loses two electrons.
PROBLEM 25.0 mL of HCl is exactly neutralised by 20.0 mL of 0.100 M NaOH. What is the concentration of the HCl?	0.08 mol/L. 1:1 ratio; c = 0.00200/0.0250 = 0.080 M.
PROBLEM A weak acid differs from a strong acid in that it	ionises only partially. Strength is about extent of ionisation.
PROBLEM What is the pH of 0.01 M NaOH at 25 °C?	12. pH = 12.
PROBLEM What are the products of an acid reacting with a metal hydroxide?	salt and water. Neutralisation.
PROBLEM Explain why a battery 'goes flat'.	A battery is a redox reaction separated into two halves. As it runs, the reactants are used up: the substance being oxidised and the one being reduced are converted to products. Once they are depleted, the electron 'push' between the two halves drops and the current stops. Recharging drives the reaction backwards with an external supply.

Module 4.8 · Organic chemistry essentials

EQUATION Write the equation: Alkane formula	C_nH_{2n+2}
EQUATION Write the equation: Esterification	$\text{acid} + \text{alcohol} \rightleftharpoons \text{ester} + \text{water}$
EQUATION Write the equation: Addition polymerisation	$n\,CH_2=CH_2 \rightarrow (CH_2-CH_2)_n$

KEY IDEA Complete the key idea: Carbon forms four …	Carbon forms four bonds and long chains
KEY IDEA Complete the key idea: Alkanes C_nH_{2n+2}; meth, …	Alkanes C_nH_{2n+2}; meth, eth, prop, but count carbons
KEY IDEA Complete the key idea: Functional groups …	Functional groups decide reactions
KEY IDEA Complete the key idea: Polymers are repeating units; …	Polymers are repeating units; proteins, starch and DNA are polymers
QUICK CHECK Carbon forms how many bonds?	4
QUICK CHECK An ester forms from	acid + alcohol
QUICK CHECK 'But-' means how many carbons?	4
QUICK CHECK Polyethylene is made from	ethene
PROBLEM What is the formula of propane (3 carbons)?	C_3H_8. C_nH_{2n+2} with n = 3.
PROBLEM Octane is an alkane with 8 carbon atoms. How many hydrogen atoms does it have?	18. 2(8) + 2 = 18.
PROBLEM Which functional group identifies an alcohol?	–OH. Hydroxyl group.
PROBLEM Vinegar's ethanoic acid belongs to which family?	carboxylic acids. –COOH group.
PROBLEM Butane and methylpropane are	isomers. Same formula, C_4H_{10}; different structures.

PROBLEM Proteins are polymers of	amino acids. Amino acids joined by peptide bonds.
PROBLEM What is the molar mass of ethanol, C_2H_5OH? (C = 12, H = 1, O = 16)	46 g/mol. 2(12) + 6(1) + 16 = 46.
PROBLEM Why can carbon form far more compounds than any other element?	Carbon forms four strong covalent bonds, so each atom can connect in several directions. Carbon–carbon bonds are strong enough to build stable long chains, branches and rings. It also bonds well with H, O, N and other elements, and can form single, double and triple bonds. Together these allow an almost limitless variety of structures.